Exp No. 5 UNDERSTANDING THE CISCO PACKET TRACER ENVIRONMENT

Aim:

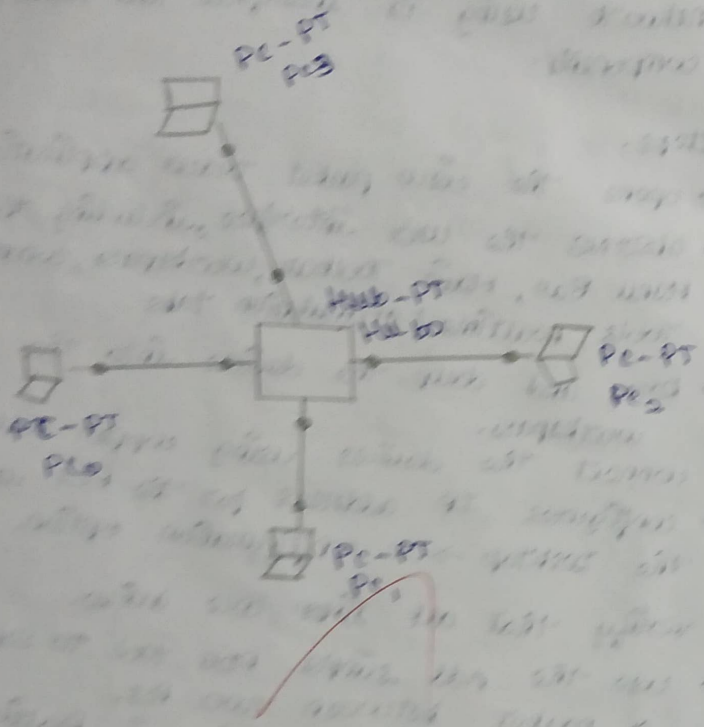
To understand the cisco packet Tracer environment and design a simple computer network using its interface and networking components.

Steps:

* opens the cisco packet tracer application.
* observe the user interface, including the Menu Bar, Main Toolbar, workspace, common Tools, realtime / simulation tabs.
* Drag and drop the devices into the workspace.
* connect the devices using copper cables.
* configures IP address for the PCs using the Desktop → IP configuration option.
* verify that all links are active.
* use the Add simple PDU tool to send a packet between two PCs.
* Observe packet transmission in realtime mode and simulation mode.
* save the network topology.

Result:

The cisco packet tracer environment was successfully explored, and a simple network was designed. the interface components, device connections, IP configuration and packet transmission were sucessfuley understood.

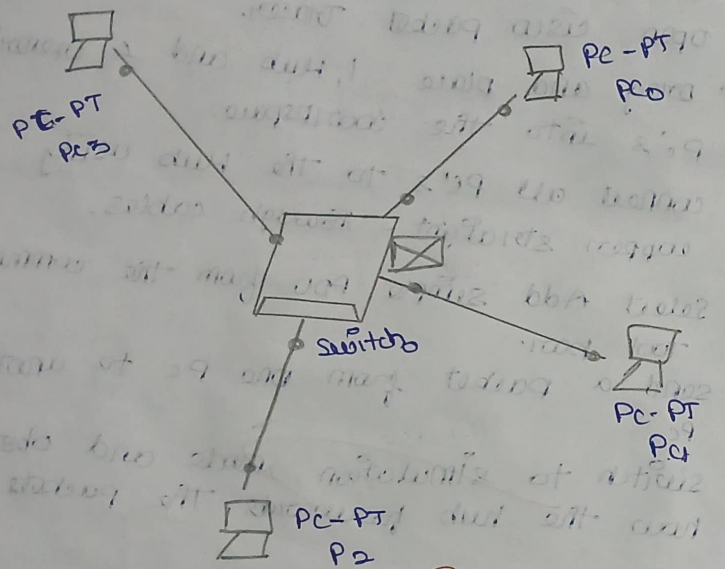


EXPERIMENT 1 : HUB

Aim:
To study the functions of a Hub by observing how it forwards data packets using this packet tracer.

Steps:
* Open this packet Tracer.
* Drag and place 1 hub and 4 generic PC's into the workspace.
* Connect all PC's to the hub using copper straight - through cables.
* Select Add simple PDU from the menu bar.
* Send a packet from one PC to another PC.
* Switch to simulation mode and observe how the hub forwards the packets.

Result:
The Hub broadcasts the received packet to all connected devices. Every PC receives the packet, but only the destination PC accepts it, while other discard it.



EXPERIMENT-2
SWITCH

Aim:

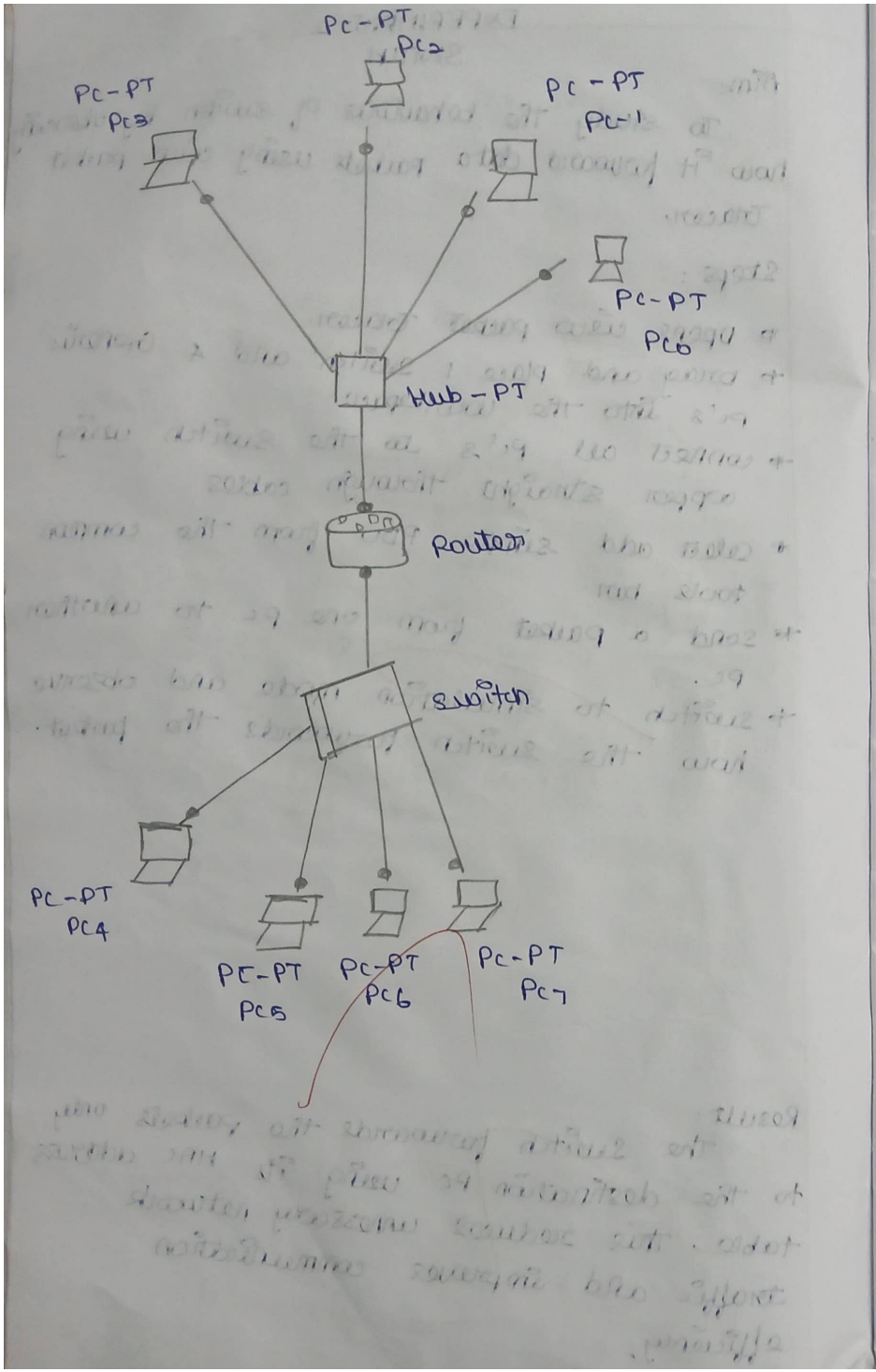
To study the behaviour of switch by observing how it forward data packets using cisco packet Tracer.

Steps:

* Open, cisco packet Tracer.
* Drag and place 1 switch and 4 Generic PC's into the workspace.
* Connect all PC's to the switch using copper straight through cables
* Select add simple PDU from the common tools bar.
* Send a packet from one PC to another PC.
* Switch to simulation mode and observe how the switch forwards the packet.

Result:

The switch forwards the packets only to the destination PC using its MAC address table. This reduces unnecessary network traffic and improves communication efficiency.



EXPERIMENT -3

HUB AND SWITCH (comparision)

Aim:

To compare the packet forwarding behaviour of a Hub and switch using cisco packet tracer.

Steps:

* create one network using a Hub and another using a switch, each connected to 4 Pc's.
* configure IP address for all Pc's in both network.
* send a simple PDU between two Pc's in each network.
* observe packet transmission.
* compare how the hub and switch forward packets.

Result:

The hub broadcast packets to all connected devices, whereas the switch forwards packets only to the intended destination. therefore, the switch provides better performance, reduce network traffic and improved efficiency compared to the hub.